Mang-yin M. Mo

mangyinm@alumni.cmu.edu | github.com/supmo668 | mmym-portfolio | cal.com/matt-mo

Objective: ML software engineer with 3+ years experience in Applied ML & deep learning applications. Early-stage contractor & engineer of multiple deep learning, agentic system based start-up in research, health-tech & ML, ex-founder of a vertical AI vision start-up,

Education

Automation Computational Science – Master of Science, Carnegie Mellon University **2021 - 2022**

- School of Computer Science, Comp Bio Dept. – GPA: 3.55

Biomedical Engineering – Bachelor of Science, Pennsylvania State University **2015 - 2018**

- School of Engineering, Biomedical Engineering Dept. – GPA: 3.55

Skills

Data Science: Scikit-learn, Pandas, Numpy, Matplotlib, OpenCV

Machine Learning: PyTorch (Lightning, TorchServe), TensorFlow (Keras), Scikit-learn, Agents (OpenAI, LangChain, Autogen), Gymnasium, ModAL

Stack, DevOps: Python-first, Typescript, Bash, Git, Docker, Cloud SDK (AWS, GCP), Kubernetes, IaaC (Terraform), 3rd party plug-ins (Prometheus, Grafana)

Work Experience

Albert-Invent

Staff Applied AI Software Engineer (LLM) **Sep 2025 - Present**

- Built LLM-driven RAG system with agentic contextual reasoning for smart vector search via **ACE (Agentic Context Engineering)** from Zou Lab @ Stanford. **ACE Paper**
- Engaged in daily software engineering responsibilities including code maintenance, consolidation, review & abstraction improvements focusing on enterprise chat services.
- Adopting best practices for prompt engineering using tracing & monitoring tools, setting up systematic RAG dashboard, metrics & datasets for scenario & pipeline study.

Healome One

ML Research Engineer **July 2024 - Sep 2025**

- Research and develop blood-based organ-specific aging clock on NHANES dataset, including data collection, curation, extraction and modelling formulation. **PyTorch, Scikit-learn, Spark**
- Design and engineer data & modelling feasibility with Cox Proportional Hazards Model, involving epigenetic & longitudinal datasets.
- Developed agentic workflows using OpenSource frameworks including Khoj & Danswer for internal function calling & GitOps. **HealomeOne**

MultiOn

ML Research Consultant **Feb 2024 - June 2024**

- Collaborated in an **Agile/Scrum** team of 8 engineers to deliver the MultiOn Chrome plug-in as a web extension & API as a product.
- Curation of data for LLM fine-tuning using **SFT** for **Llama2** LLM models on web navigation datasets and evaluation system via **LLM-as-a-Judge**. **MultiOn**

EzOut

ML Tech Lead, Co-founder **Feb 2023 - March 2024**

- Led a **Agile/Scrum** team of 2 student engineers to deploy EzOut Vision, our visual intelligence product on smart cart via **Swift** app & our fullstack **Django** & **ReactJS** service for our retail contract customers.
- Developed end-to-end **YOLOv7** pipeline from data labelling, curation & training to deployment with **Dash, TorchServe** with **Prometheus** & **Grafana** monitoring. **EzOut**

BioTrillion

CV ML Engineer **May 2022 - Sep 2022**

- Collaborated in an ML team of 5 engineers to research and develop a **PyTorch & related technologies** pipeline for eye tracking & health metric extraction.
- Created a **scikit-learn RBF kernel SVM** based unsupervised learning pipeline to discern outliers in data to implement a data QA feedback loop. **BioTrillion**

Prenetics / Rebus Biosystems

Bioengineering Research R&D Automation Associate **pre-2019 - Aug 2021**

- Performed **biological assays: PCR, ELISA, NGS, Gel assays**, **automation scripting** for **liquid handling systems, microfluidics**, maintaining GLP & scientific data analysis.
- Maintained internal data analysis pipeline for Spatial Biology assays with facilitation from **OpenCV** & **Matlab**. **Rebus**, **Prenetics**

OSS & Paper

AgentGym – Contributor **2025**

- Onboarded contributor to AgentGym, a framework for evolving generally-capable LLM-based agents across diverse environments. **GitHub**

Llama2D – Contributor **2023**

- Prepared **SFT** dataset for Llama2 fine-tuning on OCR-based RAG system for web agents, enabling multi-modal understanding of rendered DOM contents. **GitHub**, **HuggingFace**

ABGT Conference Paper – Rebus Biosystems **2021**

- Co-authored research on whole-brain spatial validation of single-cell gene expression with sub-micron resolution using spatial transcriptomics. **ResearchGate**

Technical Projects

Team Projects

Multi-Modal Scientific Notetaking Agentic System (Gary Y.) **Jan 2025**

- Benchmarked a pipeline of document agentic systems with visual interpretability to create structural notes suitable for scientific publication and analysis through Meta Ray-Ban AI Glasses. **LangGraph, Notion, AWS, CLIP, TwelveLabs & LAVIS**
- Top 5% in YC S25. **Git Pages**

Academic Projects

Zhao's Photonics Lab – Graduate Student ML Researcher **Feb 2023 - March 2024**

- Led collaborative efforts with wet-lab researchers to develop end-to-end pipeline for cancer diagnostic based on fluorescent & histology images produced in the lab. **PyTorch & computer vision**
- Benchmarked **ViTs (Vision Transformers)** against **ensemble CNNs**, with **GradCAM**-like technologies for interpretability. **Lab Site**

Generative Model Prototyping – Deep Learning Researcher **Sept 2022 - Oct 2022**

- Built **vision transformers** for bird generation (CUB-200), optimizing LSGAN/WGAN-GP architectures to achieve improvement over baseline DCGAN based on original publication.
- Designed **latent space analysis** framework evaluating disentanglement in beta-VAEs, published in course report with ablation studies across 16 hyperparameter configurations.

Intro to Cloud Computing – User Recommendation System **March 2023**

- Automated cloud infrastructure deployment using **Terraform** (EMR clusters) and **Helm/Kubernetes** (microservices), reducing provisioning time by 70% while maintaining strict cost controls ($0.70/h MySQL/$1.10/h HBase).
- Implemented CI/CD pipeline for **Docker** containers on **AWS ECR**, with **performance benchmarking** of storage tiers (HBase vs MySQL) achieving target throughput of 2000+ requests/sec.

Machine Learning Final Project – Organ-aging Clock **May 2022 - Dec 2022**

- Researched & implemented an organ-specific aging clock through **PyTorch** on **Boltzmann Machines** & **Probabilistic NN**. **Medium**

Prized Hackathons

AGI House MCP Build Day – Brand-View **May 2025**

- 1st Place for deployed MCP **Fetch.ai** solution to **Agentverse** and **ASI:One** for Brand SEO by LLM-driven search aggregation and analytics.

Google Gemini and Weights & Biases – Scientific Protocol Revisions Agent for XR **Oct 2024**

- Building on XR Portal, an **agentic system** to perform continuous revisions with **HiLT (Human-in-the-loop) feedback** from **XR Apple Vision**. **Live App**

AGI House: AIxBio Hackathon – XR Portal for Reproducible Science **Oct 2024**

- Built a **Next** client, **agentic frameworks: OpenAI, Autogen, LangChain** to allow users in Apple Vision Pro to produce actionable protocols across manual & autonomous steps and perform continuous revisions. **Git Pages**

AGI House: Agent – Llama2D **May 2024**

- Created a multi-modal agentic system for extracting & acting on rendered DOM contents. **GitHub**

OutsideLLM Hackathon Series – Mixart **June 2023**

- Created a **ReactJS** app that allows users to create remixes of music from a prompt & doodle drawing using **OpenAI**, **Bard**, & **MusicGen** on **HuggingFace**. **Live App**

Personal Projects

Aleh M. & Ge. – AI System for Market Reformulation Food Manufacturing **March 2025 - Present**

- Developed full-stack food-manufacturing agentic AI system for market & health-based reformulation. **LangChain, OpenAI, Autogen, Reflex** **Live App**

Self – FiCast: Monetize Podcast with AI Characters **Oct 2024**

- Built **NextJS** portal to allow end-to-end podcast generation using AI-powered search for AI-characters voice conversations with **Autogen**. **Live App**